Parameterizing the Genetic Architecture Under Stabilizing Selection

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Motivation

1. Figure adapted from Yengo et al. (2022).

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- In quantitative genetics, a (empirical) Bayesian linear model is used.

$$y = \sum_j G_j \beta_j + \varepsilon$$

- How do we determine the prior on β?

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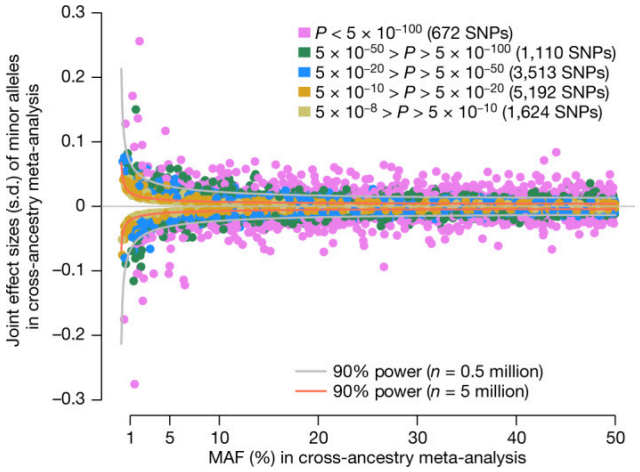
Motivation

- Across many complex traits, variants with larger effects tend to be rarer.[1]



- A standard statistical description is the α-model:

$$\mathrm{Var}(\beta_j \mid p_j) \propto [p_j(1-p_j)]^{-\alpha}$$

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Motivation

- The phenomenon is attributed to negative selection
- Variants with larger effect are removed by selection
- Many proposals, all are merely *curve fitting*

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- Introduce a focal trait y that is coupled to an unobserved selected trait z.
- Derive the induced frequency dependence of β_j.

Evolutionary setup

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- This yields a stationary allele-frequency density that depends on $\|\alpha_j\|^2/W_S$.

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  - ρ_{ab}^2 controls coupling between the focal trait and selection.

Approximate genetic architecture

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- Substituting this into the pleiotropy model gives

$$\mathbb{E}[\beta_j^2 \mid G_j] = \sigma_b^2 \left(1 - \rho_{ab}^2 \frac{2\frac{\sigma_a^2}{W_S}\hat{p}_j(1-\hat{p}_j)}{1 + 2\frac{\sigma_a^2}{W_S}\hat{p}_j(1-\hat{p}_j)}\right)$$

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- The parameters have direct evolutionary interpretations.
- In practice, the most identifiable selection-sensitive quantity is the product

$$\rho_{ab}^2 \sigma_a^2 / W_S$$

Simulation design

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- You simulate from your model so can never be wrong.

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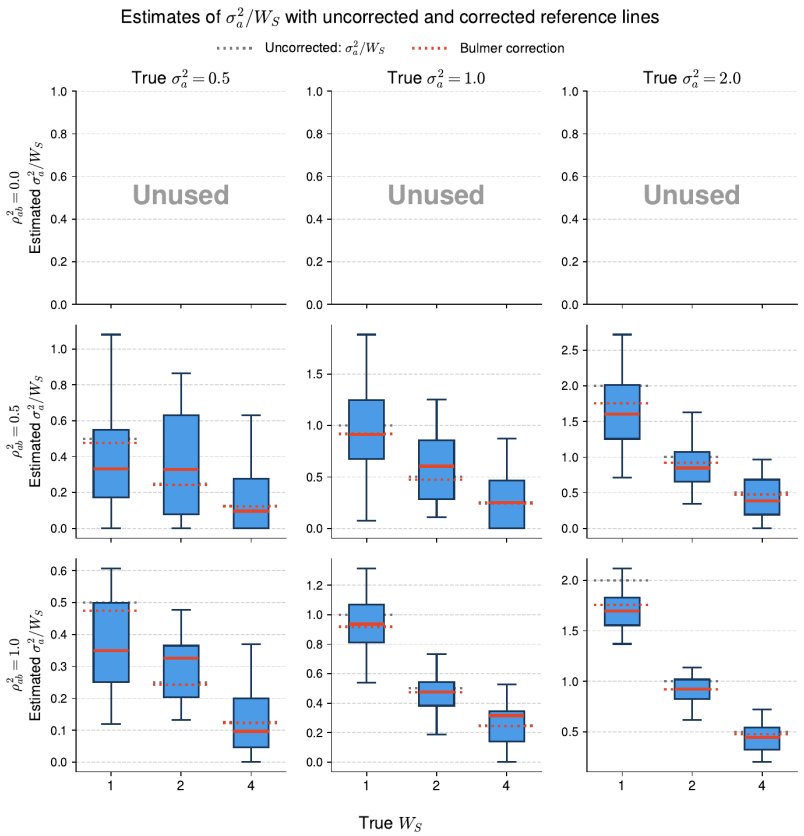
- We evaluate both variance-component recovery and held-out BLUP prediction.

Results

Recovering the selection-sensitive component



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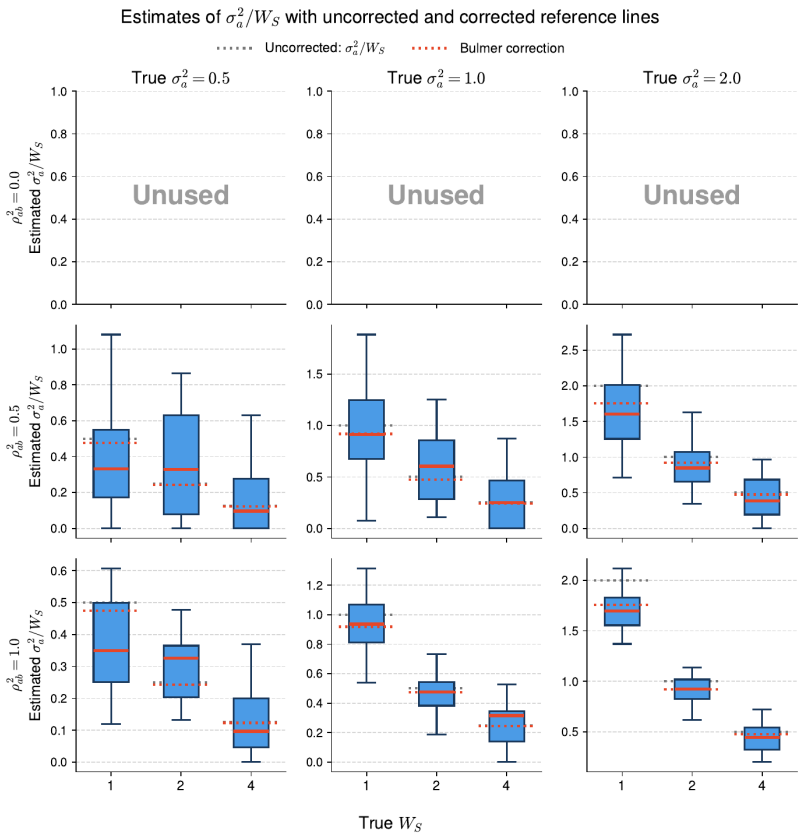

- REML broadly preserves the ordering of the true parameter settings.

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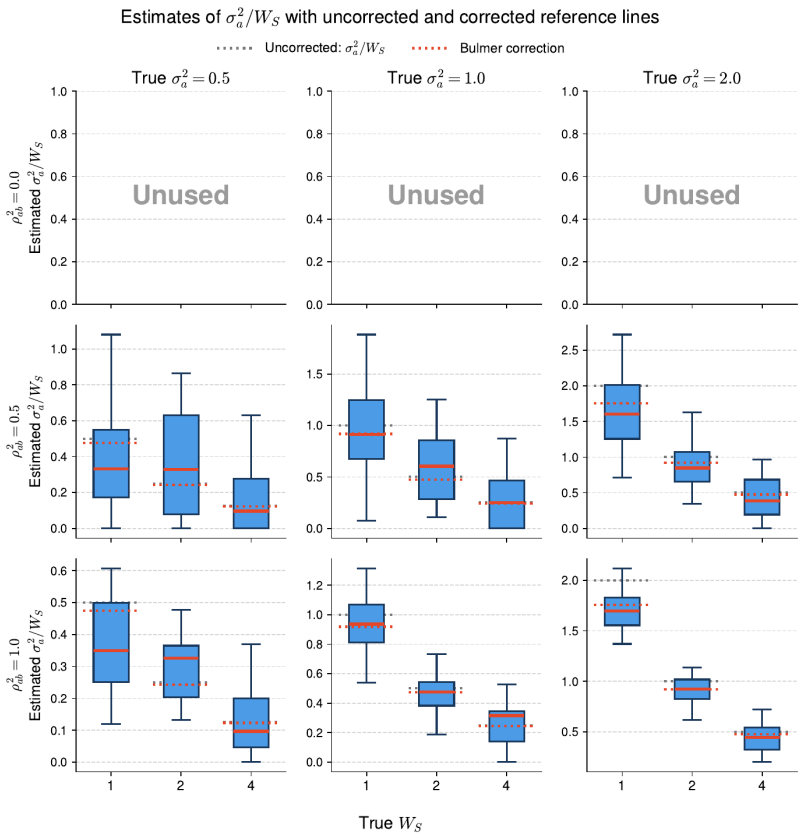

- Estimates of σ_a^2/W_S are systematically below the naive target.

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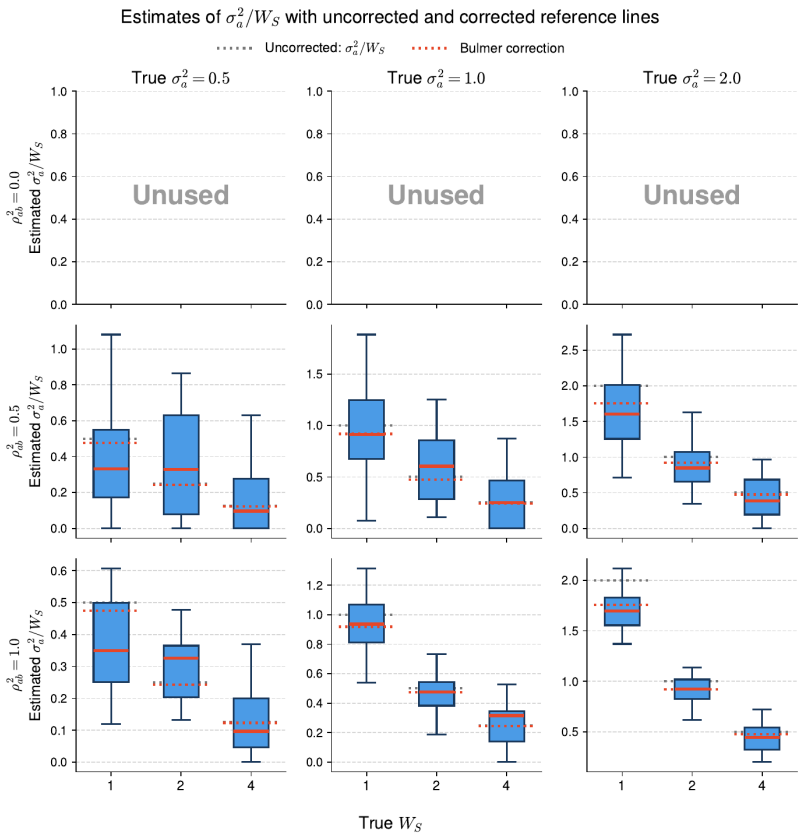
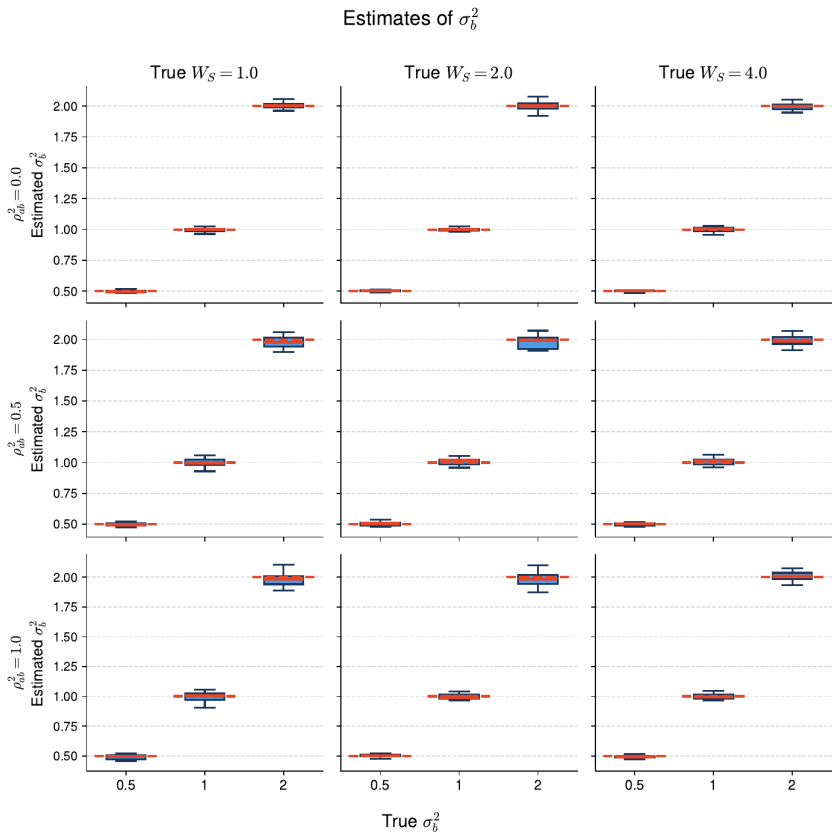


- The discrepancy is strongest where selection-induced frequency dependence is strongest, but the estimates are much closer to the Bulmer-corrected reference than to the uncorrected target.

Recovering the overall variance scale



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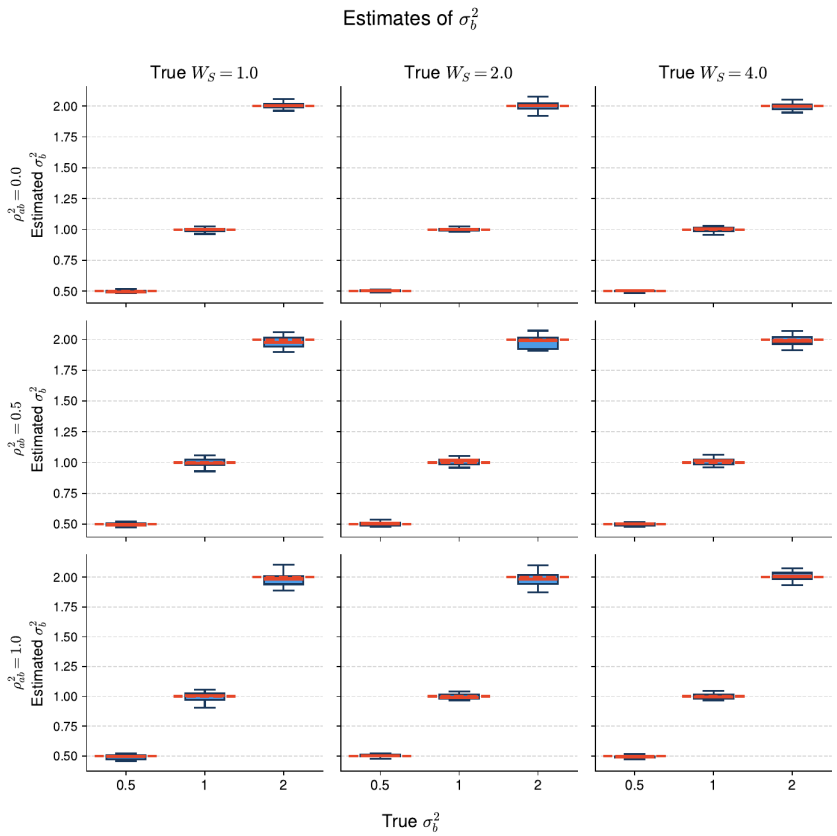

- σ_b^2 is much easier to estimate than σ_a^2/W_S.

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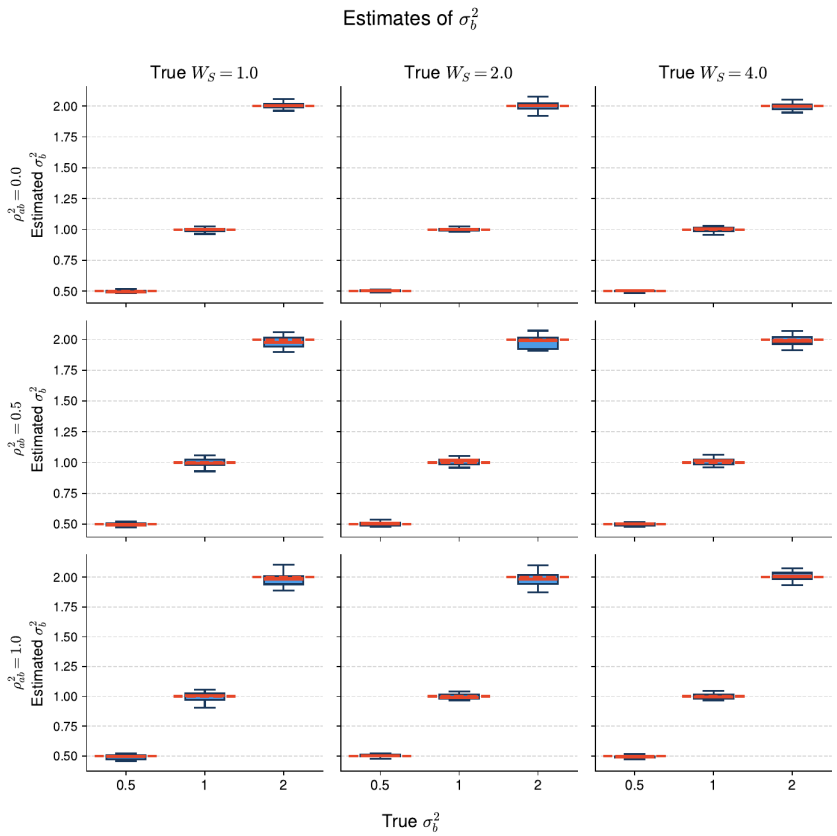

- REML estimates of σ_b^2 remain centered near the true value across the full simulation grid.

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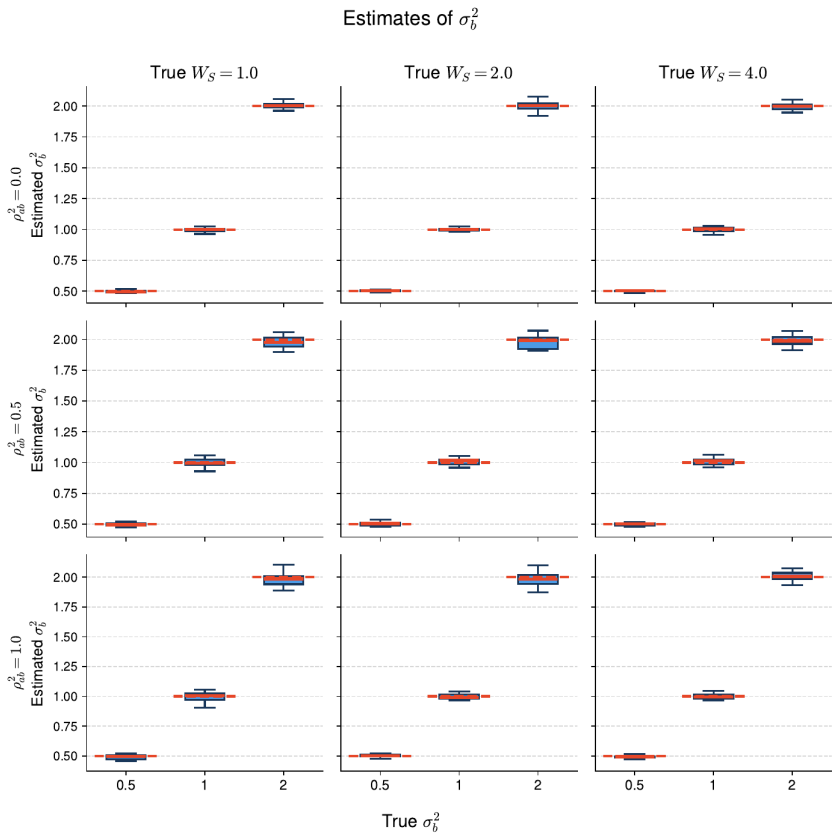
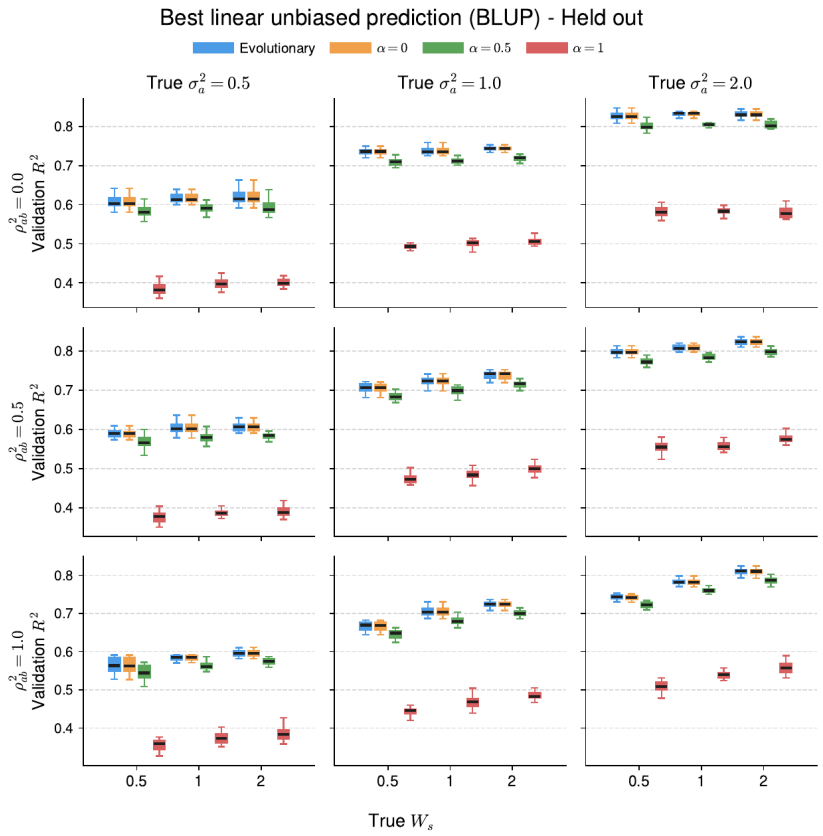


- The focal-trait variance is well identified, while the selection-sensitive component is more sensitive to departures from the single-locus approximation.

Prediction accuracy



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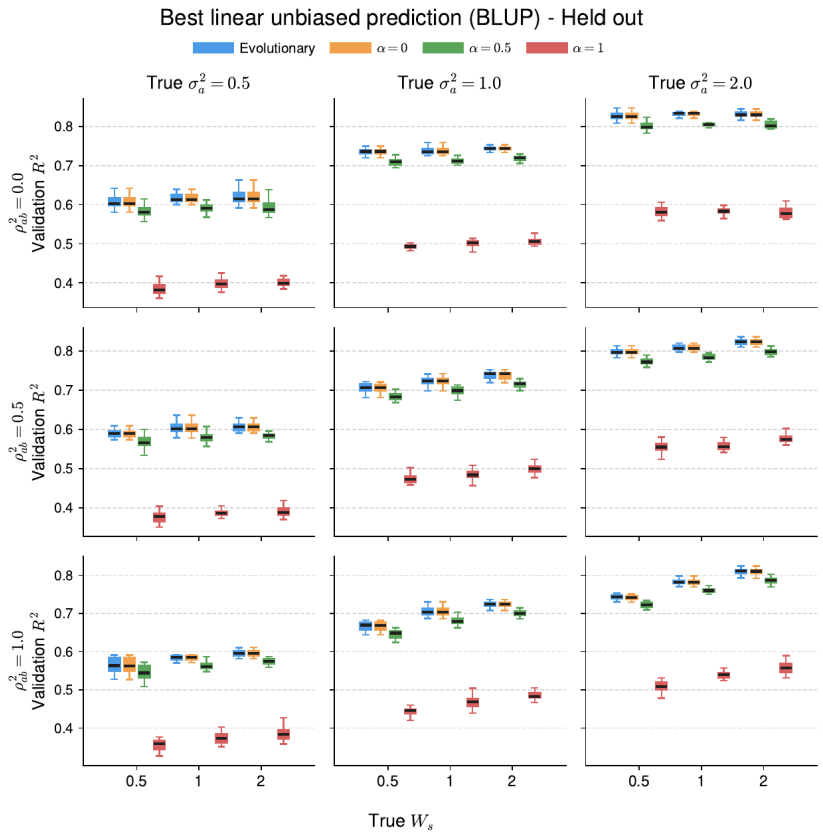

- The evolutionary model performs best overall.

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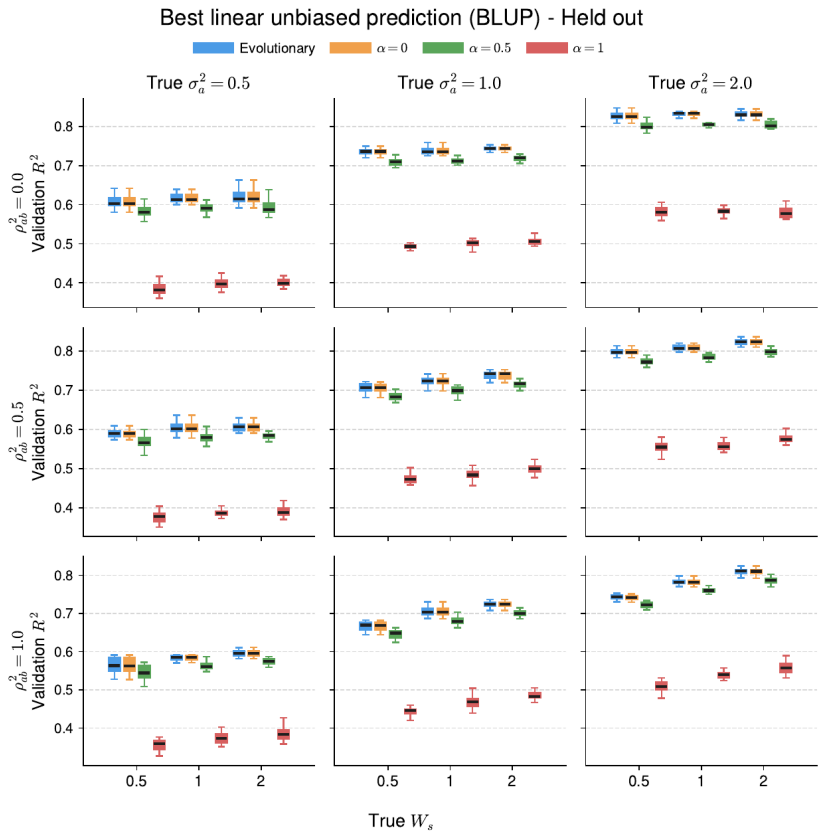
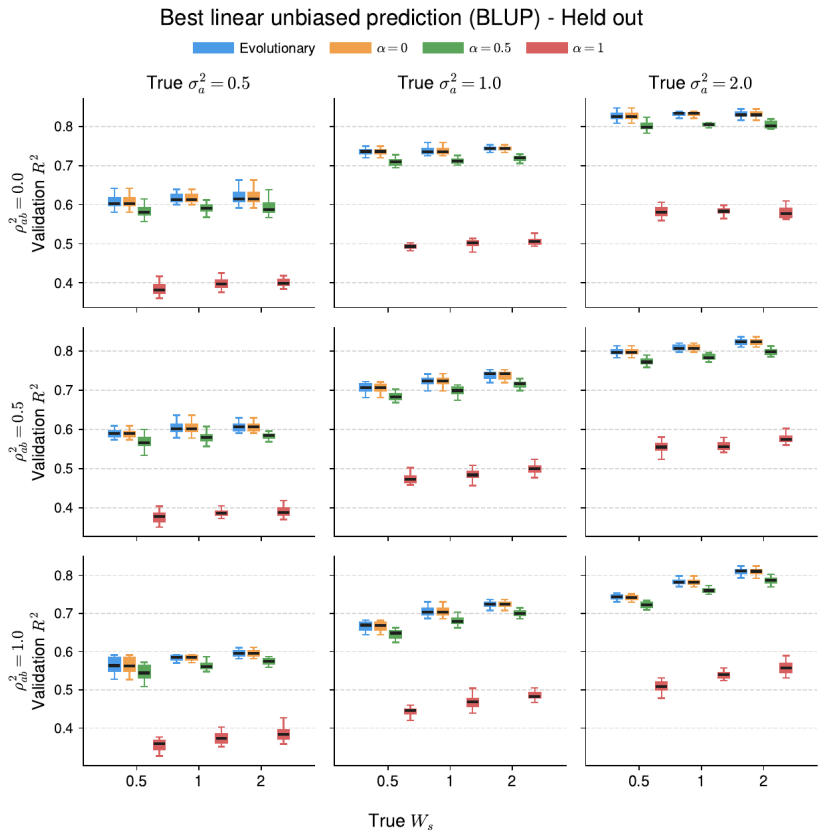


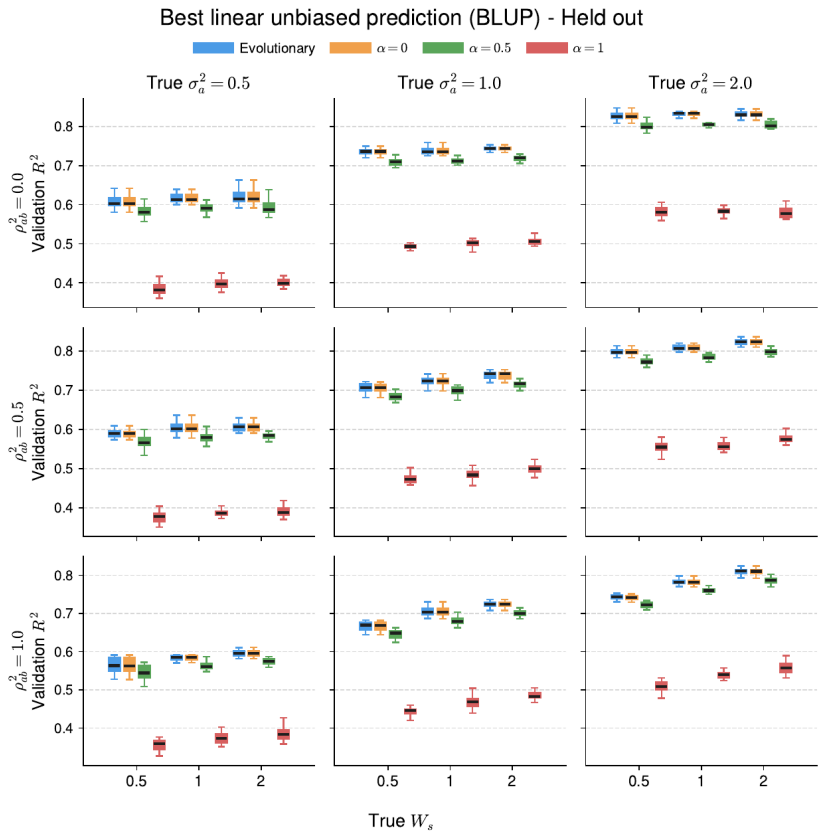
- $\alpha = 1$ performs worst throughout the grid.

Prediction accuracy



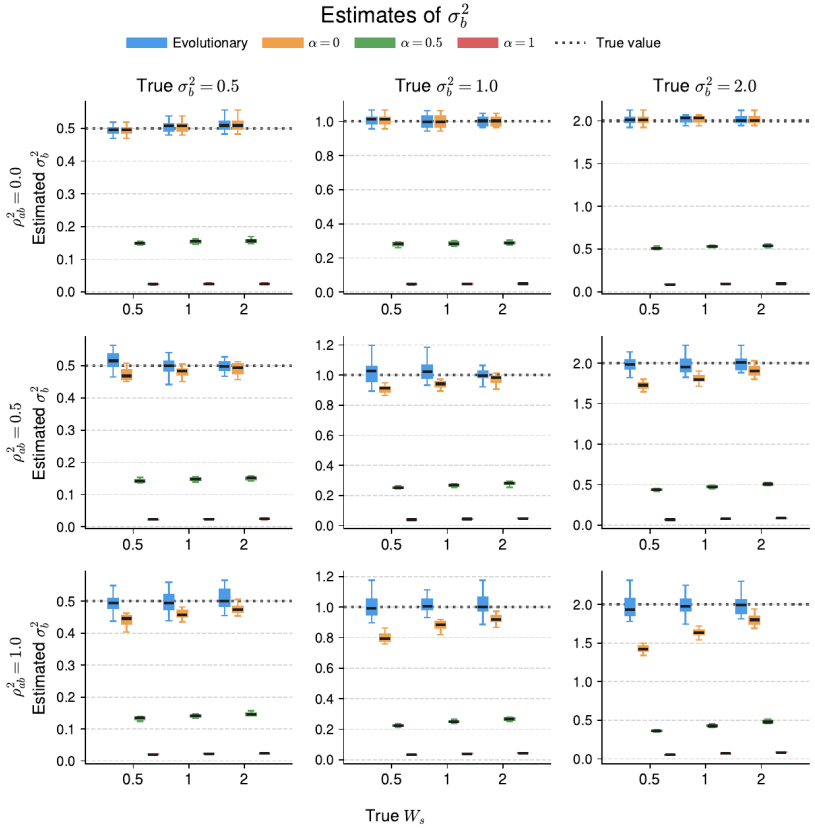
- $\alpha = 0.5$ is closer, but still remains below the evolutionary fit through most of the grid; $\alpha = 0$ is often close, especially when $\rho_{ab}^2 = 0$.

Prediction accuracy



- Similar BLUP accuracy does not imply recovery of the same architecture.

Variance components



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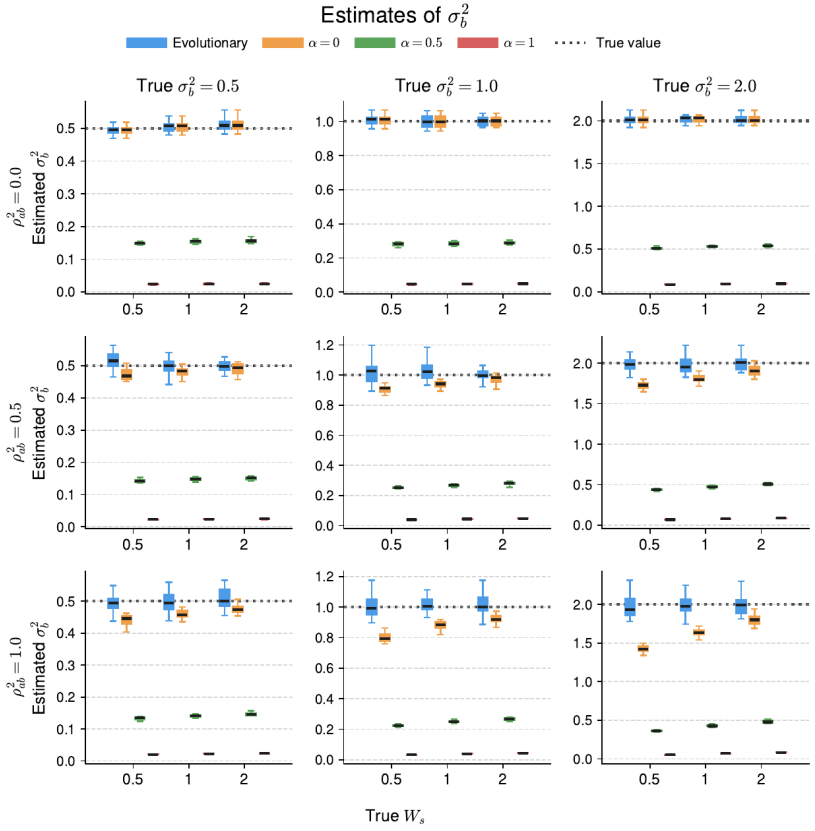

- The evolutionary model remains close to the true σ_b^2.

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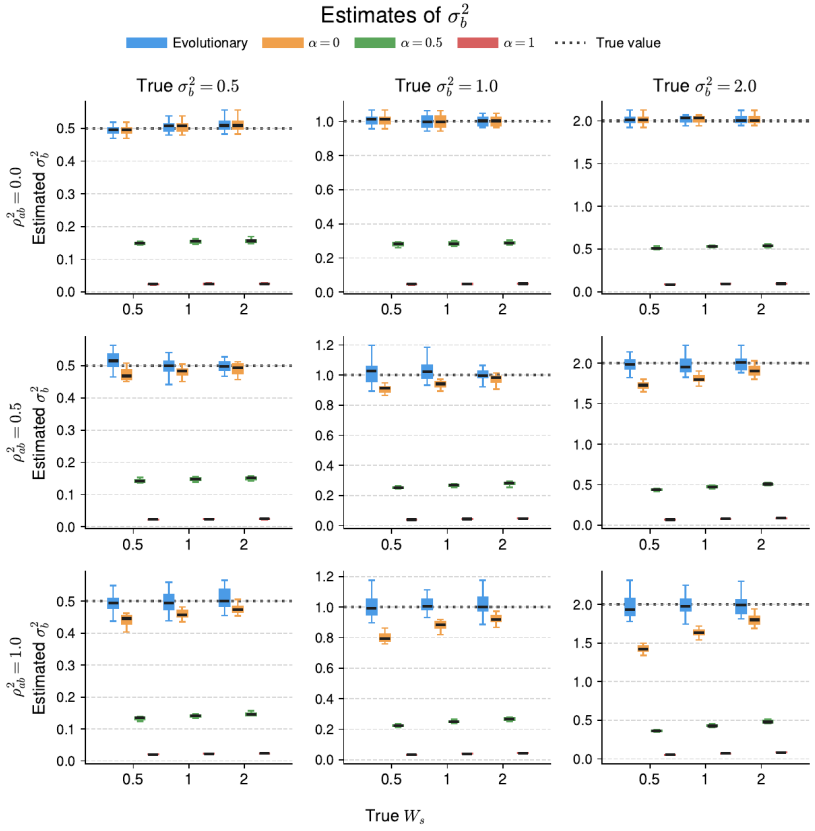

- The $\alpha = 0$ model is approximately unbiased only when $\rho_{ab}^2 = 0$ and becomes downward biased once the focal trait inherits frequency dependence.

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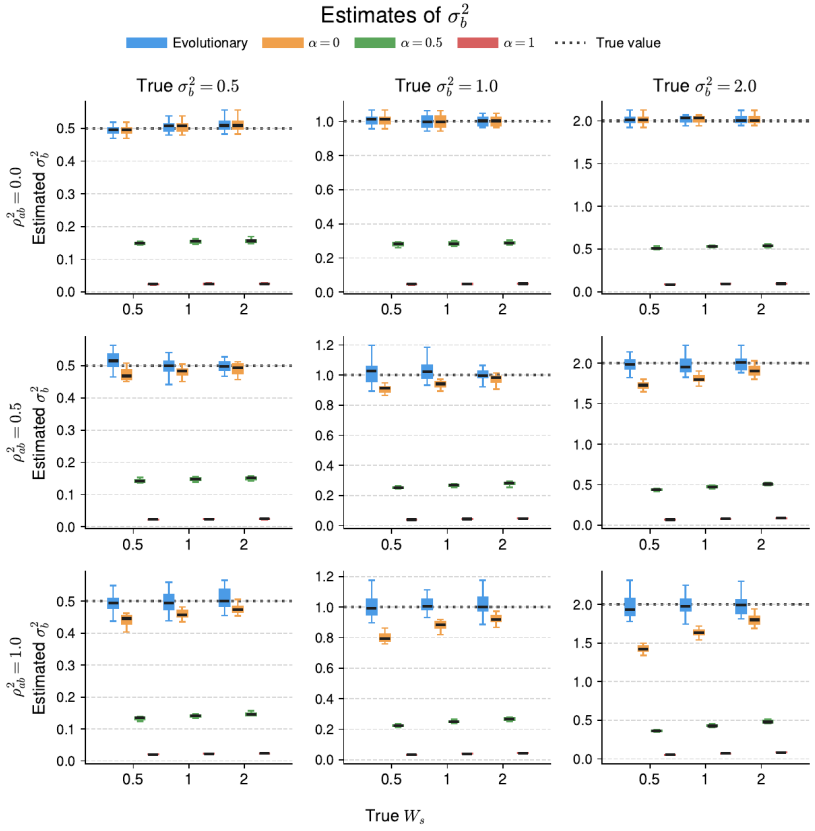

- The $\alpha = 0.5$ and $\alpha = 1$ fits underestimate σ_b^2 much more severely, showing that the α-model misses the frequency-dependent architecture.

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- We can do statistical genetics like physics instead of random curve fitting.
- In simulations, the model recovers the focal-trait variance well and generally outperforms conventional α-model baselines in prediction.
- Extending the framework to richer pleiotropy models and explicit linkage disequilibrium is a natural next step.

References

Fisher, Ronald Aylmer. 1930. *The Genetical Theory of Natural Selection*. Oxford University Press. https://openlibrary.org/books/OL6898400M/The_genetical_theory_of_natural_selection.

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